Adaptive Fuzzy Neural Network Control for a Constrained Robot Using Impedance Learning

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***Abstract*— This paper investigates adaptive fuzzy neural network (NN) control using impedance learning for a constrained robot, subject to unknown system dynamics, the effect of state constraints, and the uncertain compliant environment with which the robot comes into contact. A fuzzy NN learning algorithm is developed to identify the uncertain plant model. The prominent feature of the fuzzy NN is that there is no need to get the prior knowledge about the uncertainty and a sufficient amount of observed data. Also, impedance learning is introduced to tackle the interaction between the robot and its environment, so that the robot follows a desired destination generated by impedance learning. A barrier Lyapunov function is used to address the effect of state constraints. With the proposed control, the stability of the closed-loop system is achieved via Lyapunov's stability theory, and the tracking performance is guaranteed under the condition of state constraints and uncertainty. Some simulation studies are carried out to illustrate the effectiveness of the proposed scheme.**

***Index Terms*— Adaptive control, constraint, fuzzy logic control, impedance learning, neural networks (NN), robot.**

I. Introduction

In RECENT years, the neural network (NN) control design and stability analysis of a robot have received sufficient attention [1]–[6]. Robots' working space and the environment are usually strictly unseparated in actual life. Direct interaction between robots and environment will open new horizons of man/machine interface and change human life style completely [7]–[10]. In robotic applications, robot control must be subject to certain constraints, which typically include constraints of position and velocity. Furthermore, violation of full state constraints during operation may result in undesired performances, such as performance degradation, hazards, or system damages [11], [12]. It is therefore crucial to design an intelligent controller that has a learning ability to identify the uncertain plant model and compensate for the effect of state constraints and maintain the stability of the whole system [13], [14]. The main issue of this paper is to analyze and design an adaptive fuzzy NN controller using impedance learning algorithm for robot-environment interaction from the control point of view.

It is well known that the adaptive NN control has a learning capability [15]–[20], and has been considered as a powerful tool to identify any nonlinear function to any desired accuracy in control and applications for nonlinear systems [21]–[27]. In [28], the learning capability from the NN structure is used to identify the uncertain dynamic functions. In [29], the NN control is proposed for an MIMO nonlinear system, where the robust integral of NN and error sign control is used, so that the tracking performance improves automatically. In [30], in order to demand linguistic rules instead of learning data as prior knowledge, a fuzzy system based on the current output error and its time derivative is adopted to determine the value of control parameters. The approach has been shown to be effective in tracking a prescribed desired trajectory. However, it does not have any automatic learning capabilities to handle the uncertainty. After that, some research results have incorporated fuzzy techniques to NN structures in order to provide good overall learning capabilities [31]. Zappone *et al.* [32], Liang *et al.* [33], Gradshteyn and Ryzhik [34], and Zhou *et al.* develop robust fuzzy control approaches for a class of MIMO systems by using the learning capability of fuzzy NNs. Stochastic disturbances are considered in [36], and an adaptive fuzzy algorithm is proposed of uncertain nonlinear systems. That strategy can reduce online computation load by using few adjustable parameters. In [37], adaptive fuzzy control is used of nonstrict-feedback nonlinear systems with input saturation. In [38], fuzzy output feedback control is introduced of large-scale systems, where both unstructured uncertainties and time-varying delays are considered. In [39], fuzzy tracking control is developed of nonlinear networked systems subject to parameter uncertainties and unmeasurable state variables. Based on the research results mentioned previously, fuzzy NNs combining both NN structures and fuzzy systems, can unite advantages and exclude disadvantages. However, the approaches proposed previously are not considered the effect of constraint or the interaction with its unknown environment. Therefore, the fuzzy NN control scheme needs to be further developed for robot control subject to the effect of constraint and the robot–environment interaction.

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Position control methods give adequate performances for an uncertain robot, and only require the robotic end-effector to track a desired trajectory in free space. However, while the robotic end-effector comes in contact with the environment, it is inevitable that an interaction force will develop between the robot and its environment [40]–[42]. It is, therefore, obvious that the position control does not apply for robotic platforms when a robot is in contact with any surface. Research studies then focus on how to regulate the robot–environment interaction [43]. The main challenge in this research field is to investigate physical robot–environment interaction problems [44]. Hogan first presents impedance control to regulate the interaction between the robotic end-effector and the force exerted on the environment. Based on the research result mentioned in [45], some researchers have demonstrated the potential learning capacity to regulate the interaction between the robot and its environment. In [46], two adaptive impedance control algorithms that characterize a desired dynamic relation of the robotic end-effector with environment are proposed for robotic manipulators. As for the approaches mentioned previously, the system stability is guaranteed and good tracking performances are achieved. However, those approaches are based on the assumption that the robotic dynamics is known. In [47], impedance learning is studied for the robot interacting with unknown environments. In [48], adaptive impedance control is presented for an n-link robotic manipulator with input saturation. The authors employ radial basis function NNs to approximate the system uncertainties. However, another common issue in robot control is that the robot control must be subject to certain constraints from real-time implementation point of view.

To compensate for the effect of constraints, existing research results have demonstrated that the barrier Lyapunov function (BLF) can guarantee the constraints satisfaction [49]. In [11], an adaptive NN controller is introduced to tackle the issue with the output constraint and input deadzone. However, this method aims to handle with output constraint merely. In [50], the BLF is employed for SISO nonlinear systems to prevent constraint violation. In [51], the adaptive NN control is presented for a class of output feedback systems, where the BLF is introduced to ensure the boundedness of the closed-loop signals. Furthermore, the BLF is widely applied in flexible systems. The BLF is also proposed to suppress the undesirable vibration for a flexible string in [52]–[54].

To address the previous unsolved problem, we consider a constrained robot with unknown plant model in interactions with its environment, where the fuzzy NN structure is applied to identify the unknown plant model, and impedance learning is used to improve the robot–environment interaction. Compared with the existing work, the main contributions are highlighted as follows. First, in this paper, with respect to the traditional BLFs utilized in [50] and [52], a tan-type Lyapunov function is developed to handle the unknown state constraints, and effective fuzzy NN control is designed using impedance learning. This type of proposed control is suitable for environment interaction and objection manipulation. Next, a learning approach for robotic control systems is presented, which does not require knowledge of robotic dynamics. Unknown plant model is approximated by constructing appropriate fuzzy NN structures. There is no need to get a sufficient amount of observed data. Moreover, the stability can be ensured using the Lyapunov stability theorem. Finally, the bounds of fuzzy approximation errors are not necessarily known for control design.

II. Problem Formulations and Preliminaries

A. System Descriptions

We investigate a robot control system, which includes a constrained robot with its unknown environment, force sensors located at the robot and measure the force exerted by the environment to the robot. There is no information about the environment, and the robot dynamics is unknown. Moreover, the robot should be derived to predefined constrained space. The problem in this paper to be discussed is how to identify the unknown plant model, how to handle the effect of state constraints, and how to make the robot achieve appropriate adjustment to handle the change from its environment.

Considering the following Cartesian dynamics of an n-link robotic system [55]:

$$D_x(q)\ddot{x} + C_x(q,\dot{q})\dot{x} + G_x(q) = F_x - F_e \tag{1}$$

where $q \in \mathbb{R}^n$ is the vector of joint variables, $x \in \mathbb{R}^n$ is the vector of position of space, $D_x(q) \in \mathbb{R}^{n\times n}$ denotes the symmetric positive definite Cartesian inertia matrix, which is symmetric and positive definite, $C_x(q,\dot{q}) \in \mathbb{R}^{n\times n}$ denotes the Cartesian Coriolis and centrifugal matrix, $G_x(q) \in \mathbb{R}^n$ denotes the vector of Cartesian gravitational forces, $F_x \in \mathbb{R}^n$ is the applied control forces, $F_e \in \mathbb{R}^n$ is the force exerted by the environment. It should be noted that $D_x(q)$ and $C_x(q,\dot{q})$ should satisfy that $\dot{D}_x(q) - 2C_x(q,\dot{q})$ is skew-symmetric.

The system states $x = [x_1, x_2, \ldots, x_n]^T$ are required to satisfy the constraints as follows:

$$|x_i| < k_{ci}, \quad \forall t > 0, \ \ (i = 1, 2, \ldots, n) \tag{2}$$

where k_{ci} is a positive constant, which represents the predefined constraint on the state x_i, provided $|x_i(0)| < k_{ci}$. The output constraint is denoted as the set $\chi_1 := \{x \in \mathbb{R}^n | |x_1| < k_{c1}, \cdots, |x_n| < k_{cn}, \forall t \geq 0\} \subset \mathbb{R}^n$, and the full state constraints are denoted as the set $\chi_2 := \{x \in \mathbb{R}^n | |x_1| < k_{c1}, \cdots, |x_n| < k_{cn}, ||\dot{x}|| < k_b, \forall t \geq 0\} \subset \mathbb{R}^n$, where k_b is a positive constant.

From (1), $D_x(q)$, $C_x(q,\dot{q})$, and $G_x(q)$ cannot be obtained forehand in actual applications. Therefore, the control objective is to make the constrained robot to respond to the contact force according to impedance learning in the presence of system uncertainties. The impedance between x and F_e is usually expressed as

$$D_d(\ddot{x}_c - \ddot{x}) + C_d(\dot{x}_c - \dot{x}) + G_d(x_c - x) = F_e \tag{3}$$

where $x_c \in \mathbb{R}^n$ is the commanded trajectory, which is bounded and twice differentiable, F_e is 0 if the robot is moving in free space, and the matrices D_d, C_d, $G_d \in \mathbb{R}^{n\times n}$ are defined by the user.

We can realize the impedance control objective from the impedance model (3) if x closely tracks the desired impedance trajectory $x_d \in \mathbb{R}^n$, there is

$$D_d\ddot{x}_d + C_d\dot{x}_d + G_d x_d = -F_e + D_d\ddot{x}_c + C_d\dot{x}_c + G_d x_c \quad (4)$$

where x_d is the desired trajectory, $x_d(0) = x_c(0)$ and $\dot{x}_d(0) = \dot{x}_c(0)$. It should be noted that (4) may be interpreted as a simply filter and x_d is obtained online if x_c, D_d, C_d, G_d, and the contact force F_e are given.

B. Fuzzy Neural Networks

A fuzzy NN consists of four parts: the fuzzifier, the knowledge base, the defuzzifier, and the fuzzy inference engine working on fuzzy rules [56]. Consider l fuzzy IF-THEN rules $\mathbb{R}^{(k)}$: if x_1 is A_1^k and $\cdots$ and x_n is A_n^k, Then, y is $W^k, k = 1, \ldots, l$, where $\mathbb{R}^k$ denotes the kth rule, $1 \leq k \leq l, (x_1, x_2, \ldots, x_n)^T \in \mathbb{U} \subset \mathbb{R}^n$, and $y \in \mathbb{R}$ are the linguistic variables that are associated with the inputs and output of the fuzzy NN, respectively, and A_i^k and W^k denote the fuzzy sets in $\mathbb{U}$ and $\mathbb{R}$. The fuzzy NN performs a nonlinear mapping from $\mathbb{U}$ to $\mathbb{R}$ [57]. In this paper, the fuzzy NN is

$$g(x) = \frac{\sum_{k=1}^{l} g_k(\Pi_{i=1}^{n}\mu_{A_i^k}(x_i))}{\sum_{k=1}^{l}(\Pi_{i=1}^{n}\mu_{A_i^k}(x_i))} \quad (5)$$

where $x = [x_1, x_2, \ldots, x_n]^T$ and $\mu_{A_i^k}(x_i)$ is the membership function of linguistic variable x_i with $\mu_{A_i^k}(x_i) = \exp[-((x_i - c_{ik}^2)/(\sigma_{ik}^2))]$. For clarity, the weight vector and fuzzy basis function vector are defined, respectively, as $\vartheta = [g_1, g_2, \ldots, g_l]^T$ and $\varphi(x, c, \sigma) = [s_1, s_2, \ldots, s_l]^T$, where $s_k = (\prod_{i=1}^{n}\mu_{A_i^k}(x_i)/([\sum_{k=1}^{l}\prod_{i=1}^{n}\mu_{A_i^k}(x_i)]))$, $c = [c_1^T, c_2^T, \ldots, c_n^T]^T$ and $\sigma = [\sigma_1^T, \sigma_2^T, \ldots, \sigma_n^T]^T$. Therefore, (5) can be represented as

$$g = \vartheta^T\varphi(x, c, \sigma). \quad (6)$$

It has been proven that the fuzzy NN (6) has a learning capacity to identify any continuous functions to any degree of accuracy [58]. Therefore, we have the following approximation for functions $f_i(x_i)$, $i = 1, 2, \ldots, n$:

$$f_i(x_i) = \vartheta_i^{*T}\varphi(x_i) + \epsilon_i \quad (7)$$

where ϑ_i^{*T} is an unknown constant parameter vector, $\varphi(x_i)$ is the fuzzy basis function and ϵ_i is the approximation error, which satisfies $\max_{Z\in\Omega_Z}||\epsilon_i|| < \epsilon_i^*$, where $\epsilon_i^* > 0$ is unknown bound.

C. Preliminaries

In this paper, a BLF is presented to tackle the constraint problems. A *tan*-type BLF is introduced, modified from the one presented in [59], as follows:

$$V = \frac{k_b^2}{\pi}\tan\left(\frac{\pi\phi_1^2}{2k_b^2}\right), \quad ||\phi_1(0)|| < k_b \quad (8)$$

where ϕ_1 is the state, and $k_b > 0$ is the constraint on the state ϕ_1.

Remark 1: By L' Hospital's rule, we get

$$\lim_{k_b\to\infty}\frac{k_b^2}{\pi}\tan\left(\frac{\pi\phi_1^2}{2k_b^2}\right) = \frac{1}{2}\phi_1^2 \quad (9)$$

which means if there is no constraint requirement, we can simply replace the BLF with the quadratic form. Therefore, the *tan*-type BLF analysis for robot systems is a general approach.

Lemma 1 [60]: Let function $V(t) > 0$ be a continuous function defined $\forall t \in \mathbb{R}^+$ and $V(0)$ bounded. If the following inequality holds: $\dot{V}(t) \leq -c_1V(t) + c_2$, where c_1 and c_2 are positive constants, then we can conclude that $V(t)$ is bounded.

The Nussbaum gain technique is used in this paper. A function $N(\zeta)$ is called a Nussbaum-type function satisfying [61]

$$\lim_{s\to\infty}\sup\frac{1}{s}\int_0^s N(\zeta)d\zeta = +\infty \quad (10)$$

$$\lim_{s\to\infty}\inf\frac{1}{s}\int_0^s N(\zeta)d\zeta = -\infty. \quad (11)$$

Commonly used Nussbaum functions include: $k^2\cos(k)$, $k^2\sin(k)$, $k\cos(k)$, and $k\sin(k)$ [62]. In this paper, $k\cos(k)$ is exploited. The following Lemma will be employed in the control design.

Lemma 2: $\zeta(\cdot)$ and $V(\cdot)$ are smooth functions defined on $[0, t_f)$ with $V(t) > 0$, $\forall t \in [0, t_f)$. $N(\zeta) = \zeta * cos((\pi/(2))\zeta)$ is an smooth Nussbaum-type function. If the following inequation holds:

$$V(t) \leq c_0e^{-c_1t} + e^{-c_1t}\int_0^t g_1(t)N(\zeta)\dot{\zeta}e^{c_1\tau}d\tau \quad (12)$$

$\forall t \in [0, t_f)$, where c_1 is a positive constant, $g_1(t)$ is a time-varying parameter, which takes values in the unknown closed intervals $I_1 := [l_1^-, l_1^+]$ with $0 \notin I_1$, and c_0 represents a suitable constant, then $V(t)$, $\zeta(t)$, and $\int_0^t g_1(t)N(\zeta)\dot{\zeta}d\tau$ must be bounded on $[0, t_f)$.

Proof. See Appendix.

III. Control Design

For the dynamic system (1), it is nontrivial to build a control scheme to handle the effect of constraints. The problem is further complicated to tackle all state constraints. In this paper, control schemes are proposed to solve control design with output constraint and full state constraints, respectively. Let $x_1 = x$ and $x_2 = \dot{x}$, the Cartesian error signals are defined as

$$z_1 = x_1 - x_d \quad (13)$$

$$z_2 = x_2 - \alpha_1 \quad (14)$$

where $z_1 = [z_{11}, \ldots, z_{1n}]^T$, $z_2 = [z_{21}, \ldots, z_{2n}]^T$, and α_1 is the virtual control input. Fig. 1 shows the system structure.

A. Control Design With Output Constraint

In this paper, according to (1), the robotic dynamics can therefore be expressed as

$$\begin{aligned}\dot{x}_1 &= x_2\\ \dot{x}_2 &= D_x^{-1}(q)[F_x - F_e - C_x(q,\dot{q})x_2 - G_x(q)] \quad (15)\\ y &= x_1.\end{aligned}$$

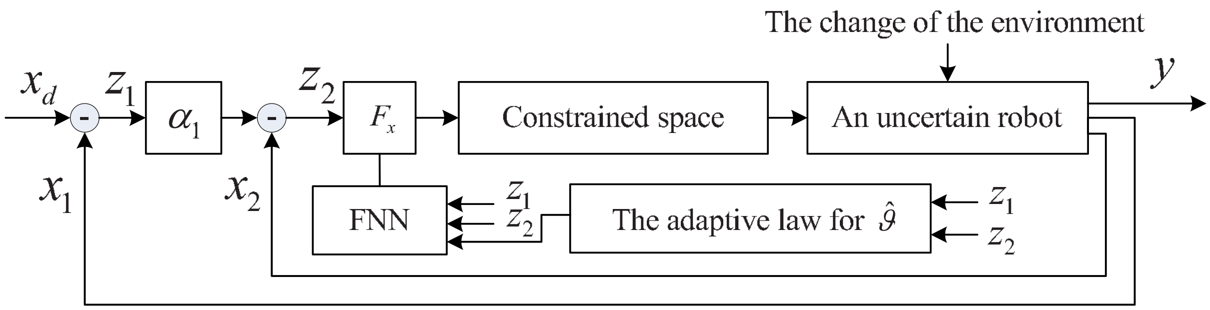

Fig. 1. System structure.

Assume that the error signal z_1 should be required to satisfy the constraints as $|z_{1i}| < k_{ai}$, $(i = 1, 2, \ldots, n)$. The virtual control α_1 in (14) is designed as

$$\alpha_1 = -K_1 \begin{bmatrix} \frac{k_{a1}^2}{z_{11}\pi} \sin\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)\cos\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right) \\ \cdots \\ \frac{k_{an}^2}{z_{1n}\pi} \sin\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)\cos\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right) \end{bmatrix} + \dot{x}_d \quad (16)$$

where $K_1 = \text{diag}[k_{11}, \ldots, k_{1n}]$ is a positive constant design matrix.

Remark 2: By L' Hospital's rule, there is

$$\lim_{x\to 0} \frac{1}{x} \sin\left(\frac{\pi x^2}{2k_a^2}\right)\cos\left(\frac{\pi x^2}{2k_a^2}\right) \to 0. \quad (17)$$

where k_a is a positive constant. Therefore, singularity for this term will not occur in (16).

Taking the contact force F_e into consideration, adaptive control is modified, supposing the dynamics is completed known, by adding the contact force F_e, the model-based control is

$$F_x^* = -\begin{bmatrix} \frac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \frac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} + D_x(q)\dot{\alpha}_1 + C_x(q, \dot{q})\alpha_1 + G_x(q) - K_p z_2 + F_e, \quad (18)$$

where $K_p = \text{diag}[k_{p1}, k_{p2}, \ldots, k_{pn}]$ is a positive constant design matrix. Substituting (18) into (1), the error dynamics can be obtained as

$$D_x\dot{z}_2 + C_x z_2 = -\begin{bmatrix} \frac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \frac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} - K_p z_2. \quad (19)$$

Considering the following Lyapunov function candidate as:

$$V_1(t) = \sum_{i=0}^{n} \frac{k_{ai}^2}{\pi} \tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) + \frac{1}{2} z_2^T D_x z_2. \quad (20)$$

Taking the time derivative of $V_1(t)$, results in

$$\dot{V}_1(t) = -\sum_{i=0}^{n} k_{1i} \frac{k_{ai}^2}{\pi} \tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) - z_2^T K_p z_2 \le -\rho_1 V_1(t) \quad (21)$$

where

$$\rho_1 = \min\left(2k_{11}, 2k_{12}\ldots, 2k_{1n}, \frac{2\lambda_{\min}(K_p)}{\lambda_{\max}(D_x)}\right) \quad (22)$$

with $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ denote the maximum and the minimum eigenvalue of $(\cdot)$, respectively. To ensure the closed-loop system stability, control parameters should be set as $k_{1i} > 0 (i = 1, 2, \ldots, n)$ and $K_p > 0$. So, $\dot{V}_1(t)$ will be negative definite. However, since the uncertainties exist in $D_x(q)$, $C_x(q, \dot{q})$, and $G_x(q)$, the model-based control (18) may be not be realizable. Therefore, fuzzy NN is used to approximate the uncertainties via the online learning. The adaptive controller is given as

$$F_x = -\begin{bmatrix} \frac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \frac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} + \hat{\vartheta}_D^T \varphi_D \dot{\alpha}_1 + \hat{\vartheta}_C^T \varphi_C \alpha_1 + \hat{\vartheta}_G^T \varphi_G - K_p z_2 + F_e - K_r \text{sgn}(z_2) \quad (23)$$

where $K_r = \text{diag}[k_{rii}]$ is dimensionally compatible matrix, unknown constant parameter vectors ϑ_D, ϑ_C, and ϑ_G are fuzzy NN weights, $\hat{\vartheta}_D$, $\hat{\vartheta}_C$, and $\hat{\vartheta}_G$ are the estimates of ϑ_D, ϑ_C, and ϑ_G, respectively. The adaptive functions are designed as

$$\dot{\hat{\vartheta}}_{Di} = -\varphi_{Di}\dot{\alpha}_1 z_{2i} - \sigma_{Di}\hat{\vartheta}_{Di} \quad (24)$$
$$\dot{\hat{\vartheta}}_{Ci} = -\varphi_{Ci}\alpha_1 z_{2i} - \sigma_{Ci}\hat{\vartheta}_{Ci} \quad (25)$$
$$\dot{\hat{\vartheta}}_{Gi} = -\varphi_{Gi} z_{2i} - \sigma_{Gi}\hat{\vartheta}_{Gi} \quad (26)$$

where σ_{Di}, σ_{Ci}, σ_{Gi}, $(i = 1, 2, \ldots, n)$ are small position constants for improving the robustness [63]. The fuzzy NN $\hat{\vartheta}_D^T \varphi_D$ is an approximation of $\vartheta_D^{*T} \varphi_D$, $\hat{\vartheta}_C^T \varphi_C$ is an approximation of $\vartheta_C^{*T} \varphi_C$ and $\hat{\vartheta}_G^T \varphi_G$ is an approximation of $\vartheta_G^{*T} \varphi_G$, and

$$\vartheta_D^{*T} \varphi_D = D_x(q) + \epsilon_D \quad (27)$$
$$\vartheta_C^{*T} \varphi_C = C_x(q, \dot{q}) + \epsilon_C \quad (28)$$
$$\vartheta_G^{*T} \varphi_G = G_x(q) + \epsilon_G \quad (29)$$

where ϵ_D, ϵ_C, and ϵ_G are approximation errors of fuzzy NNs. Let $\tilde{(\cdot)} = \hat{(\cdot)} - (\cdot)^*$. Substituting the controller (23) into (1), the error dynamics is obtained as

$$D_x\dot{z}_2 + C_x z_2 = -\begin{bmatrix} \dfrac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \dfrac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} + \tilde{\vartheta}_D^T\varphi_D\dot{\alpha}_1 + \tilde{\vartheta}_C^T\varphi_C\alpha_1 + \tilde{\vartheta}_G^T\varphi_G - K_p z_2 + E_x - K_r\mathrm{sgn}(z_2) \quad (30)$$

where $E_x = \epsilon_D\dot{\alpha}_1 + \epsilon_c\alpha_1 + \epsilon_G$. Considering the Lyapunov function candidate as

$$V_2(t) = V_1(t) + \frac{1}{2}\tilde{\vartheta}_D^T\tilde{\vartheta}_D + \frac{1}{2}\tilde{\vartheta}_C^T\tilde{\vartheta}_C + \frac{1}{2}\tilde{\vartheta}_G^T\tilde{\vartheta}_G. \quad (31)$$

The time derivative of $V_2(t)$ is

$$\begin{aligned}\dot{V}_2(t) = {} & \sum_{i=0}^{n}\frac{z_{1i}z_{2i}}{\cos^2\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right)} - \sum_{i=0}^{n}k_{1i}\frac{k_{ai}^2}{\pi}\tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) \\ & + \frac{1}{2}z_2^T\dot{D}_x z_2 - z_2^T C_x z_2 - z_2^T K_p z_2 \\ & + z_2^T(E_x - K_r\mathrm{sgn}(z_2)) - \sum_{i=0}^{n}\frac{z_{1i}z_{2i}}{\cos^2\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right)} \\ & + z_2^T\tilde{\vartheta}_D^T\varphi_D\dot{\alpha}_1 + z_2^T\tilde{\vartheta}_C^T\varphi_C\alpha_1 + z_2^T\tilde{\vartheta}_G^T\varphi_G \\ & + \tilde{\vartheta}_D^T\dot{\hat{\vartheta}}_D + \tilde{\vartheta}_C^T\dot{\hat{\vartheta}}_C + \tilde{\vartheta}_G^T\dot{\hat{\vartheta}}_G. \end{aligned} \quad (32)$$

Noting that

$$z_2^T\tilde{\vartheta}_D^T\varphi_D\dot{\alpha}_1 = \sum_{k=1}^{n}\tilde{\vartheta}_{Dk}^T\varphi_{Dk}\dot{\alpha}_1 z_{2k} \quad (33)$$

$$z_2^T\tilde{\vartheta}_C^T\varphi_C\alpha_1 = \sum_{k=1}^{n}\tilde{\vartheta}_{Ck}^T\varphi_{Ck}\alpha_1 z_{2k} \quad (34)$$

$$z_2^T\tilde{\vartheta}_G^T\varphi_G = \sum_{k=1}^{n}\tilde{\vartheta}_{Gk}^T\varphi_{Gk}z_{2k}. \quad (35)$$

Substituting (24)–(26) into (32) and the gain of the term K_r, is chosen as $k_{rii} > ||E_{xi}||$, there is

$$\dot{V}_2(t) = -\sum_{i=0}^{n}k_{1i}\frac{k_{ai}^2}{\pi}\tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) - z_2^T K_p z_2 - \sigma_D\tilde{\vartheta}_D^T\hat{\vartheta}_D - \sigma_C\tilde{\vartheta}_C^T\hat{\vartheta}_C - \sigma_G\tilde{\vartheta}_G^T\hat{\vartheta}_G. \quad (36)$$

Since $-\tilde{\vartheta}_i^T\hat{\vartheta}_i = -\tilde{\vartheta}_i^T(\vartheta_i^* + \tilde{\vartheta}_i) = -\tilde{\vartheta}_i^T\tilde{\vartheta}_i - \tilde{\vartheta}_i^T\vartheta_i^*$ and $-\tilde{\vartheta}_i^T\vartheta_i^* \le (1/(2))(\tilde{\vartheta}_i^T\tilde{\vartheta}_i + \vartheta_i^{*T}\vartheta_i^*)$, there is $-\tilde{\vartheta}_i^T\hat{\vartheta}_i \le -(1/(2))\tilde{\vartheta}_i^T\tilde{\vartheta}_i + (1/(2))\vartheta_i^{*T}\vartheta_i^*$. Substituting inequalities $-\tilde{\vartheta}_D^T\hat{\vartheta}_D \le -(1/(2))\tilde{\vartheta}_D^T\tilde{\vartheta}_D + (1/(2))\vartheta_D^{*T}\vartheta_D^*$, $-\tilde{\vartheta}_C^T\hat{\vartheta}_C \le -(1/(2))\tilde{\vartheta}_C^T\tilde{\vartheta}_C + (1/(2))\vartheta_C^{*T}\vartheta_C^*$ and $-\tilde{\vartheta}_G^T\hat{\vartheta}_G \le -(1/(2))\tilde{\vartheta}_G^T\tilde{\vartheta}_G + (1/(2))\vartheta_G^{*T}\vartheta_G^*$ into (36), results in

$$\begin{aligned}\dot{V}_2(t) \le {} & -\sum_{i=0}^{n}k_{1i}\frac{k_{ai}^2}{\pi}\tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) - z_2^T K_p z_2 - \frac{1}{2}\tilde{\vartheta}_D^T\tilde{\vartheta}_D \\ & - \frac{1}{2}\tilde{\vartheta}_C^T\tilde{\vartheta}_C - \frac{1}{2}\tilde{\vartheta}_G^T\tilde{\vartheta}_G + \frac{1}{2}\left(\vartheta_D^{*T}\vartheta_D^* + \vartheta_C^{*T}\vartheta_C^*\vartheta_G^{*T}\vartheta_G^*\right) \\ \le {} & -\rho_2 V_2(t) + c_2 \end{aligned} \quad (37)$$

where

$$\rho_2 = \min\left(2k_{11}, 2k_{12}\ldots, 2k_{1n}, \frac{2\lambda_{\min}(K_p)}{\lambda_{\max}(D_x)}\right) \quad (38)$$

$$c_2 = \frac{1}{2}\left(\vartheta_D^{*T}\vartheta_D^* + \vartheta_C^{*T}\vartheta_C^* + \vartheta_G^{*T}\vartheta_G^*\right). \quad (39)$$

To ensure the closed-loop system stability, control parameters $k_{1i} > 0(i = 1, 2, \ldots, n)$ and $K_p > 0$. Therefore, we have Theorem 1.

Theorem 1: For system dynamics described by (15) with output constraint, and the control (23) with the adaptation laws (24)–(26), given that the initial conditions are bounded, it can be concluded that the target impedance is achieved and the errors will eventually converge to a small neighborhood around zero by appropriately choosing design parameters. Furthermore, the tracking error z_1 converges asymptotically to the compact set $\Omega_{z_1} := \left\{z_1 \in \mathbb{R}^2|||z_1 \le \sqrt{B_1}||\right\}$ where $B_1 = 2(V_2^*(0) + \frac{c_2}{\rho_2})$ with ρ_2 and c_2 given in (38) and (39). The proof of Theorem 1 is given in the Appendix.

B. Control Design With All State Constraints

The adaptive control using learning capability from the fuzzy NN structure with full state constraints will be presented in this section. It should be noted that although the control design is similar to the control (23) due to the use of backstepping design and fuzzy approximations, the system states should be required to be constrained in the control design. In this sense, the control problem is more challenging than before. The error signal z_2 should be constrained satisfying $\|z_2\| < k_b$, and the detailed design process is described as follows.

According to the Moore–Penrose inverse, there is

$$z_2^T(z_2^T)^+ = \begin{cases} 0, & z_2 = [0, 0, \ldots, 0]^T \\ 1, & \text{otherwise.} \end{cases} \quad (40)$$

The model-based control is designed as

$$F_x^* = -\begin{bmatrix} \dfrac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \dfrac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} - \left(z_2^T\right)^+ K_2\frac{k_b^2}{\pi}\tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) + D_x(q)\dot{\alpha}_1 + C_x(q,\dot{q})\alpha_1 + G_x(q) - K_p z_2 + F_e \quad (41)$$

where $K_2 = \mathrm{diag}[k_{21}, k_{22}, \ldots, k_{2n}]$ is a positive constant design matrix. Substituting (41) into (1), the error dynamics is

$$D_x\dot{z}_2 = -\begin{bmatrix} \dfrac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \dfrac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} - (z_2^T)^+ K_2\frac{k_b^2}{\pi}\tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) - K_p z_2 - C_x z_2. \quad (42)$$

Considering the following Lyapunov function candidate as

$$V_3(t) = \sum_{i=0}^{n} \frac{k_{ai}^2}{\pi} \tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) + \frac{k_b^2}{\pi} \tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) + \frac{1}{2} z_2^T D_x z_2. \tag{43}$$

It should be noted that the time derivative of $V_3(t)$ will be $\dot{V}_3(t) \leq -\sum_{i=0}^{n} k_{1i}(k_{ai}^2/(\pi)) \tan((\pi z_{1i}^2/(2k_{ai}^2)))$, when $z_2 = [0, 0, \ldots, 0]^T$. Then, the stability of the system can still be drawn by Barbalat's lemma in [64]. Otherwise, in the case of $z_2 \neq [0, 0, \ldots, 0]^T$, the time derivative of $V_3(t)$ will be

$$\begin{aligned}\dot{V}_3(t) &= -\sum_{i=0}^{n} k_{1i} \frac{k_{ai}^2}{\pi} \tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) - K_2 \frac{k_b^2}{\pi} \tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) \\ &\quad - z_2^T K_p z_2 + \frac{z_2^T \dot{z}_2}{\cos^2\left(\frac{\pi z_2^T z_2}{2k_b^2}\right)} \\ &\leq -\rho_3 V_3(t) + \frac{z_2^T \dot{z}_2}{\cos^2\left(\frac{\pi z_2^T z_2}{2k_b^2}\right)} \end{aligned} \tag{44}$$

where

$$\rho_3 = \min\left(2k_{11}, 2k_{12} \ldots, 2k_{1n}, 2k_{21}, 2k_{22} \ldots, 2k_{2n}, \frac{2\lambda_{\min}(K_p)}{\lambda_{\max}(D_x)}\right). \tag{45}$$

Multiplying (44) by $e^{\rho_3 t}$ yields

$$e^{\rho_3 t} \dot{V}_3(t) \leq -\rho_3 e^{\rho_3 t} V_3(t) + e^{\rho_3 t} g_1(t) N(z_2) \dot{z}_2 \tag{46}$$

where $g_1(t) = 1/(\cos^2(\pi z_2^T z_2/(2k_b^2)))\ (\sum_{i=1}^{n} \cos(\pi z_{2i}/(2k_b^2)))$ and $N(z_2) = z_2^T \sum_{i=1}^{n} \cos(\pi z_{2i}/(2k_b^2))$.Then, integrating (46) over $[0, t_f)$, results in

$$V_3(t) \leq e^{-\rho_3 t} V_3(0) + e^{-\rho_3 t} \int_0^t g_1(\tau) N(z_2) \dot{z}_2 e^{\rho_3 \tau} d\tau. \tag{47}$$

where $t \in [0, t_f)$. According to Lemma 2, it can be concluded that $V_3(t)$, z_1, and z_2 are all bounded on $[0, t_f)$ if $\rho_3 > 0$. However, since the uncertainties exist in $D_x(q)$, $C_x(q, \dot{q})$, and $G_x(q)$, the model-based control (41) may be not be realizable. Thus, we employ the learning ability from the fuzzy NN structure to identify the uncertainties. The fuzzy NN controller is presented as

$$\begin{aligned}F_x &= -\begin{bmatrix} \dfrac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \dfrac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} - (z_2^T)^+ K_2 \frac{k_b^2}{\pi} \tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) \\ &\quad + \hat{\vartheta}_D^T \varphi_D \dot{\alpha}_1 + \hat{\vartheta}_C^T \varphi_C \alpha_1 + \hat{\vartheta}_G^T \varphi_G - K_p z_2 + F_e \\ &\quad - K_r \operatorname{sgn}(z_2) \end{aligned} \tag{48}$$

where $K_r = \operatorname{diag}[k_{rii}]$ is dimensionally compatible matrix. Substituting the controller (48) into (1), there is

$$\begin{aligned}D_x \dot{z}_2 &= -\begin{bmatrix} \dfrac{z_{11}}{\cos^2\left(\frac{\pi z_{11}^2}{2k_{a1}^2}\right)} \\ \cdots \\ \dfrac{z_{1n}}{\cos^2\left(\frac{\pi z_{1n}^2}{2k_{an}^2}\right)} \end{bmatrix} - \left(z_2^T\right)^+ K_2 \frac{k_b^2}{\pi} \tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) \\ &\quad + \tilde{\vartheta}_D^T \varphi_D \dot{\alpha}_1 + \tilde{\vartheta}_C^T \varphi_C \alpha_1 + \tilde{\vartheta}_G^T \varphi_G - K_p z_2 + E_x \\ &\quad - C_x z_2 - K_r \operatorname{sgn}(z_2) \end{aligned} \tag{49}$$

where $E_x = \epsilon_D \dot{\alpha}_1 + \epsilon_c \alpha_1 + \epsilon_G$. Considering the Lyapunov function candidate as

$$V_4(t) = V_3(t) + \frac{1}{2} \tilde{\vartheta}_D^T \tilde{\vartheta}_D + \frac{1}{2} \tilde{\vartheta}_C^T \tilde{\vartheta}_C + \frac{1}{2} \tilde{\vartheta}_G^T \tilde{\vartheta}_G. \tag{50}$$

By using $K_{rii} \geq ||E_{xi}||$ and substituting (24)–(26), the time derivative of $V_4(t)$ is

$$\begin{aligned}\dot{V}_4(t) &= \sum_{i=0}^{n} \frac{z_{1i} z_{2i}}{\cos^2\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right)} - \sum_{i=0}^{n} k_{1i} \frac{k_{ai}^2}{\pi} \tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) \\ &\quad + \frac{z_2^T \dot{z}_2}{\cos^2\left(\frac{\pi z_2^2}{2k_b^2}\right)} + \frac{1}{2} z_2^T \dot{D}_x z_2 - z_2^T C_x z_2 \\ &\quad - K_2 \frac{k_b^2}{\pi} \tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) - \sum_{i=0}^{n} \frac{z_{1i} z_{2i}}{\cos^2\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right)} \\ &\quad + z_2^T \tilde{\vartheta}_D^T \varphi_D \dot{\alpha}_1 + z_2^T \tilde{\vartheta}_C^T \varphi_C \alpha_1 + z_2^T \tilde{\vartheta}_G^T \varphi_G \\ &\quad + \tilde{\vartheta}_D^T \dot{\tilde{\vartheta}}_D + \tilde{\vartheta}_C^T \dot{\tilde{\vartheta}}_C + \tilde{\vartheta}_G^T \dot{\tilde{\vartheta}}_G - z_2^T K_p z_2. \end{aligned} \tag{51}$$

Substituting $-\tilde{\vartheta}_D^T \hat{\vartheta}_D \leq -(1/(2)) \tilde{\vartheta}_D^T \tilde{\vartheta}_D + (1/(2)) \vartheta_D^{*T} \vartheta_D^*$, $-\tilde{\vartheta}_D^T \hat{\vartheta}_D \leq -(1/(2)) \tilde{\vartheta}_C^T \tilde{\vartheta}_C + \frac{1}{2} \vartheta_C^{*T} \vartheta_C^*$, and $-\tilde{\vartheta}_G^T \hat{\vartheta}_G \leq -(1/(2)) \tilde{\vartheta}_G^T \tilde{\vartheta}_G + (1/(2)) \vartheta_G^{*T} \vartheta_G^*$, results in

$$\begin{aligned}\dot{V}_4(t) &\leq -\sum_{i=0}^{n} k_{1i} \frac{k_{ai}^2}{\pi} \tan\left(\frac{\pi z_{1i}^2}{2k_{ai}^2}\right) - K_2 \frac{k_b^2}{\pi} \tan\left(\frac{\pi z_2^T z_2}{2k_b^2}\right) \\ &\quad - \frac{1}{2} \tilde{\vartheta}_D^T \tilde{\vartheta}_D - \frac{1}{2} \tilde{\vartheta}_C^T \tilde{\vartheta}_C - \frac{1}{2} \tilde{\vartheta}_G^T \tilde{\vartheta}_G + \frac{z_2^T \dot{z}_2}{\cos^2(\frac{\pi z_2^T z_2}{2k_b^2})} \\ &\quad + \frac{1}{2} (\vartheta_D^{*T} \vartheta_D^* + \vartheta_C^{*T} \vartheta_C^* + \vartheta_G^{*T} \vartheta_G^*) - z_2^T K_p z_2 \\ &\leq -\rho_4 V_4(t) + c_4 + \frac{z_2^T \dot{z}_2}{\cos^2(\frac{\pi z_2^T z_2}{2k_b^2})} \end{aligned} \tag{52}$$

where

$$\rho_4 = \min\left(2k_{11}, 2k_{12} \cdots, 2k_{1n}, \frac{2 * \lambda_{\min}(K_p)}{\lambda_{\max}(D_x)}\right) \tag{53}$$

$$c_4 = \frac{1}{2}\left(\vartheta_D^{*T} \vartheta_D^* + \vartheta_C^{*T} \vartheta_C^* + \vartheta_G^{*T} \vartheta_G^*\right). \tag{54}$$

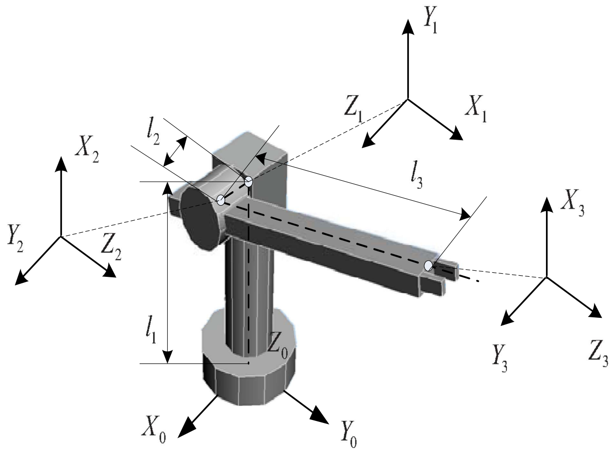

Fig. 2. Simulation scenario.

TABLE I
PARAMETERS OF THE ROBOT

Parameter	Description	Value
m_1	Mass of link 1	2.00 kg
m_2	Mass of link 2	1.00 kg
m_3	Mass of link 3	0.30 kg
l_1	Length of link 1	1.00 m
l_2	Length of link 2	0.20 m
l_3	Length of link 3	1.00 m
I_1	Inertia of link 1	00.5×10^{-3} kgm^2
I_2	Inertia of link 2	00.1×10^{-3} kgm^2

Multiplying (52) by $e^{\rho_4 t}$ and then integrating over $[0, t_f)$, there is

$$V_4(t) \le e^{-\rho_4 t}\int_0^t g_1(\tau)N(z_2)\dot{z}_2 e^{\rho_4\tau}d\tau + \frac{c_4}{\rho_4} + e^{-\rho_4 t}\left(V_4(0) - \frac{c_4}{\rho_4}\right). \tag{55}$$

It can be concluded that $V_4(t)$, z_1, z_2, $\int_0^t g_1(\tau)N(z_2)\dot{z}_2 d\tau$ are all bounded on $[0, t_f)$ according to Lemma 2 when $\rho_4 > 0$ and $c_4 > 0$. Therefore, we get Theorem 2.

Theorem 2: For system dynamics described by (15) with all state constraints, and the control (48) with the adaptation laws (24)–(26), given that the initial conditions are bounded, it can be concluded that the target impedance is achieved and the errors will eventually converge to a small neighborhood around zero by appropriately choosing design parameters. Furthermore, tracking error z_1 converges asymptotically to the compact set $\Omega_{z_1} := \left\{z_1 \in \mathbb{R}^2 || |z_1 \le \sqrt{B_2}||\right\}$ where $B_2 = 2(V_4(0) + (c_4/(\rho_4)) + M)$ with ρ_4 and c_4 given in (53) and (54), and M satisfying $\| \int_0^t g_1(\tau)N(z_2)\dot{z}_2 d\tau\| \le M$. The proof of Theorem 2 is given in the Appendix.

IV. APPLICATION STUDIES

In this section, we investigate a typical robot system defined by (1) with two rotary degrees and one prismatic degree of freedom as shown in Fig. 2, which includes a force sensor that is located at the end-effector.

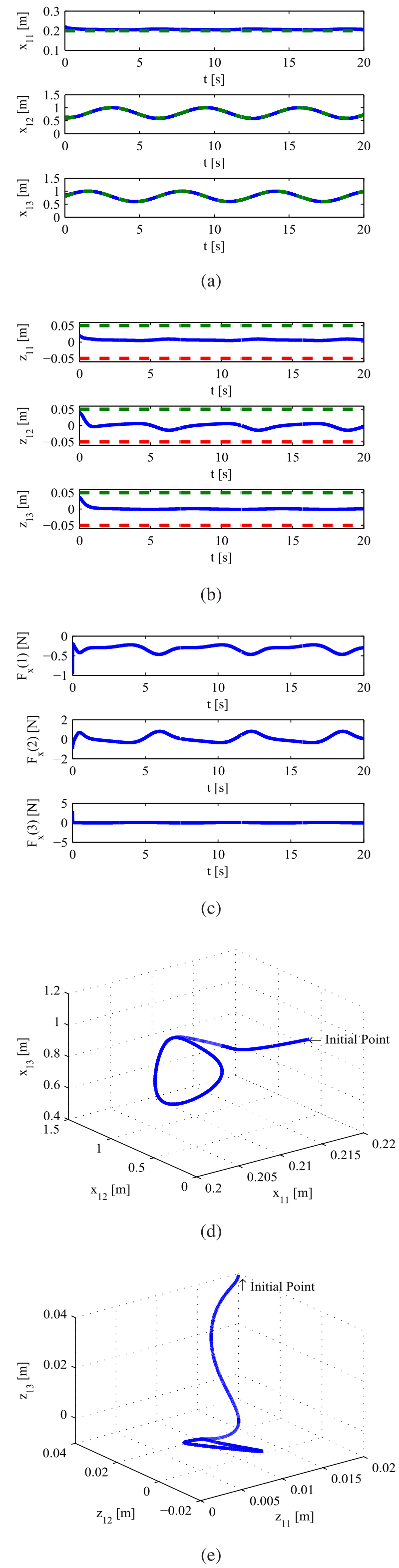

Fig. 3. Simulation results for Case 1: output constraint in free space. (a) x_1 (solid line: actual; dashed line: reference). (b) z_1 (solid line: actual; solid line: the constraint of z_1). (c) F_x. (d) Output signals in Cartesian space. (e) Error signals in Cartesian space.

Four simulation cases are carried out to verify the effectiveness of the proposed control (23) and (48) for a 3-DOF robotic manipulator with constraints, respectively. To verify that the proposed control has a capability to handle the effect

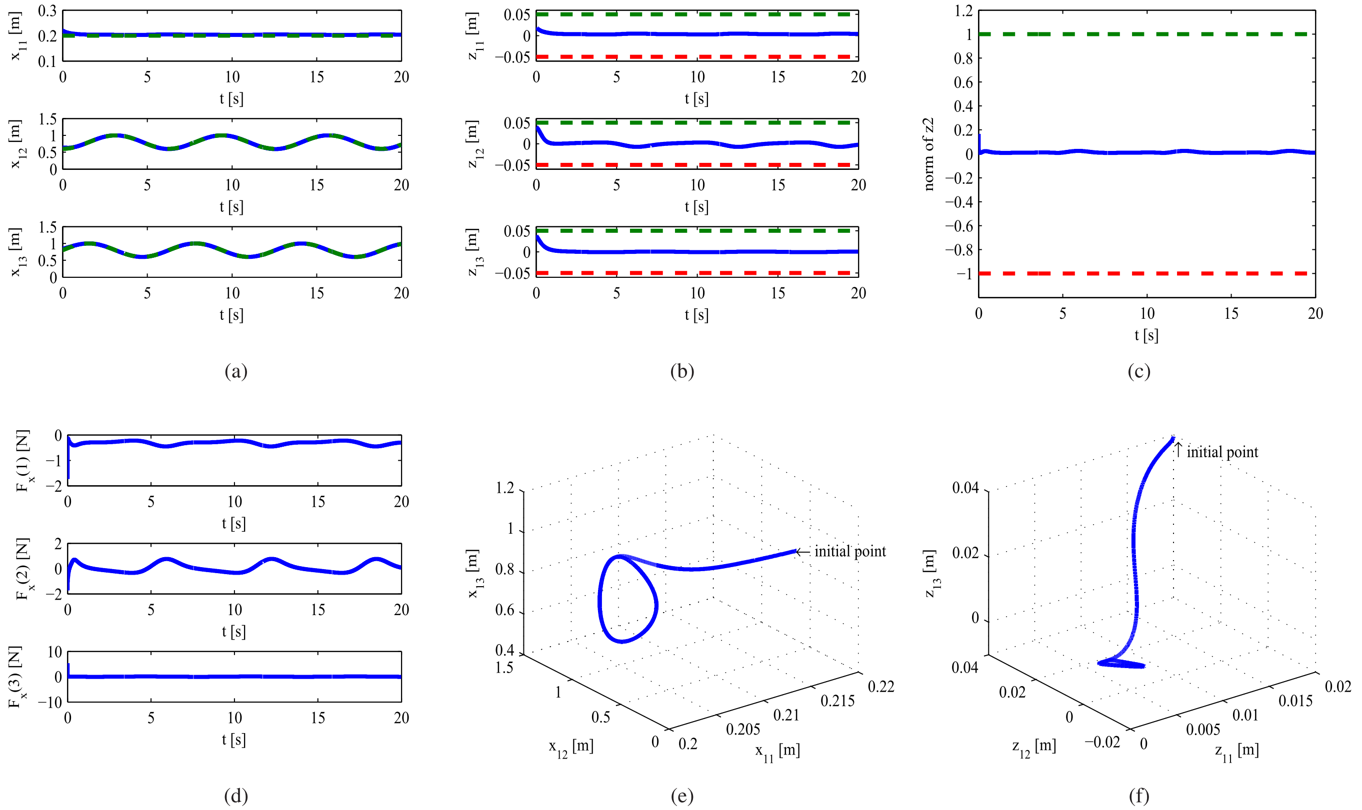

Fig. 4. Simulation results for Case 2: state constraints in free space. (a) x_1 (solid line: actual; dashed line: reference). (b) z_1 (solid line: actual; solid line: the constraint of z_1). (c) z_2 (solid line: actual; solid line: the constraint of z_2). (d) F_x. (e) Output signals in Cartesian space. (f) Error signals in Cartesian space.

of constraint in free space, the robot is required to move in free space with output constraint in Case 1 and state constraints in Case 2. After that, to verify that the proposed control has an ability to not only handle the effect of constraint, but also tackle the interaction between the robot and its environment, the robot is required to keep in contact with a wall, that is located at $Y_o = 0.8$ m, with output constraint in Case 3 and state constraints in Case 4. According to the force detected by the sensor, the robot generates appropriate control input that derives the robot to a desired destination. As for all the cases, there is no information about the environment and the unknown plant dynamics. The aim of this section is to examine whether the proposed control we designed is able to make output $x_1 = [x_{11}, x_{12}, x_{13}]^T$ follow the commanded trajectory with constraint in free space, and handle the change of unknown environment while the robot is in contact with the wall.

The matrices $D_x(q)$, $C_x(q, \dot{q})$, and $G_x(q)$ in (1) are defined as

$$D_x(q) = J^{-T}(q)D(q)J^{-1}(q) \tag{56}$$
$$C_x(q, \dot{q}) = J^{-T}(q)(C(q, \dot{q}) - DJ^{-1}(q)\dot{J}(q))J^{-1}(q) \tag{57}$$
$$G_x(q) = J^{-T}(q)G(q). \tag{58}$$

It can be seen that $D_x(q)$, $C_x(q, \dot{q})$, and $G_x(q)$ are functions of q_1 and q_2. $D(q)$, $C(q, \dot{q})$, and $G(q)$ are defined as

$$D(q) = \begin{bmatrix} m_3 q_3^2 \sin^2(q_2) + p_1 & p_2 q_3 \cos(q_2) & p_2 \sin(q_2) \\ p_2 q_3 \cos(q_2) & m_3 q_3^2 + I_2 & 0 \\ p_2 \sin(q_2) & 0 & m_3 \end{bmatrix}$$

$$C(q, \dot{q}) = \begin{bmatrix} p_4 \dot{q}_2 + p_5 \dot{q}_3 & p_4 \dot{q}_1 - p_3 q_3 p_8 & p_5 \dot{q}_1 - p_3 p_6 q_3 \\ -p_4 \dot{q}_1 & m_3 q_3 \dot{q}_3 & p_3 p_9 - m_3 q_3 \dot{q}_2 \\ -p_5 \dot{q}_1 + p_3 p_{10} & m_3 q_3 \dot{q}_2 + p_3 p_{11} & 0 \end{bmatrix}$$

$$G(q) = \begin{bmatrix} 0 \\ -m_3 g q_3 \cos(q_2) \\ -m_3 g \sin(q_2) \end{bmatrix}$$

where $p_1 = m_3 l_2^2 + m_2 l_1^2 + I_1$, $p_2 = m_3 l_2$, $p_3 = m_3 l_1$, $p_4 = m_3 q_3^2 \sin(q_2)\cos(q_2)$, $p_5 = m_3 q_3^2 \sin^2(q_2)$, $p_6 = \sin(q_2)\dot{q}_2$, $p_7 = \sin(q_2)\dot{q}_3$, $p_8 = p_6 + p_7$, $p_9 = \cos(q_2)\dot{q}_1$, $p_{10} = \cos(q_2)\dot{q}_2$, and $p_{11} = \cos(q_2)\dot{q}_3$. Parameters of the robot are defined in Table I.

The commanded trajectory is described by

$$x_c(t) = \begin{bmatrix} x_{c1} \\ x_{c2} \\ x_{c3} \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.8 - 0.2\cos(t) \\ 0.8 + 0.2\sin(t) \end{bmatrix} \tag{59}$$

which represents a circle of radius 0.2 m and its center is located at $x = [0.2, 0.8, 0.8]^T$ m. The initial position is $x(0) = [0.22, 0.64, 0.84]^T$ m, $\dot{x}(0) = [0.0, 0.0, 0.0]^T$ m/s.

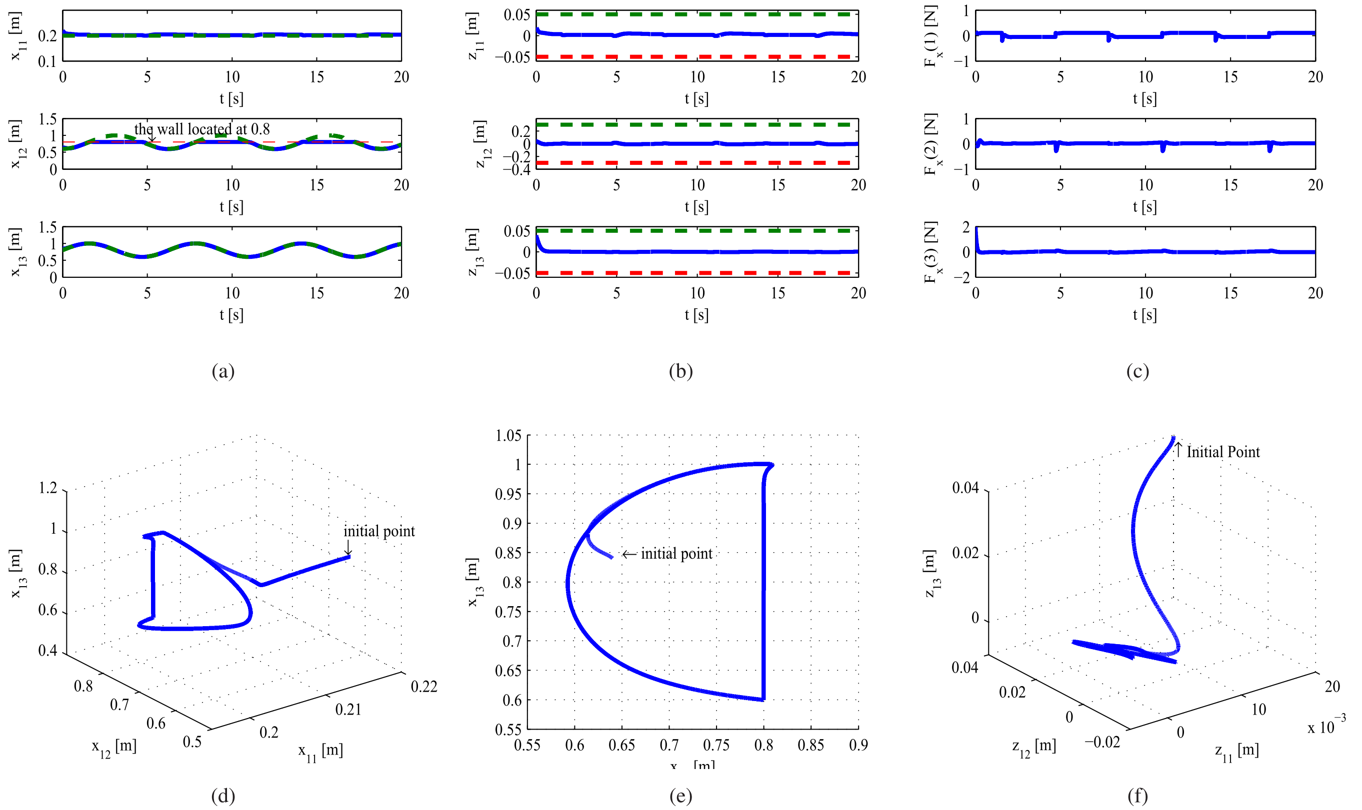

Fig. 5. Simulation results for Case 3: output constraint in constrained space. (a) x_1 (solid line: actual; dashed line: reference). (b) z_1 (solid line: actual; solid line: the constraint of z_1). (c) F_x. (d) Output signals in Cartesian space. (e) Output signals $x_{12} - x_{13}$ in Cartesian space. (f) Error signals in Cartesian space.

Case 1 *(Output Constraint in Free Space):* The robot is required to move in free space. The frontier of the output constraint is $k_a = [0.05, 0.05, 0.05]^T$. The parameters of the target impedance are chosen as $D_d = \text{diag}[0.1]$, $C_d = \text{diag}[20.0]$, and $G_d = \text{diag}[100.0]$. The gains of the control laws are chosen as $K_p = \text{diag}[20.0]$ and $K_1 = \text{diag}[5.0]$.

The simulation results are shown in Fig. 3 with the proposed controller (23). Fig. 3(a) shows that the tracking objective is achieved with output constraint, and the proposed control (23) has a nice tracking performance in free space. The tracking errors are plotted in Fig. 3(b). It is obvious that the tracking error z_1 is always in the predefined constraint, and converges to a small value near zero. The control signal is given in Fig. 3(c). The motion of the end-effector in Cartesian space is plotted in Fig. 3(d), which illustrates that the tracking performance is smooth and the proposed control (23) has an ability to handle the effect of output constraint. Fig. 3(e) gives the error signal in Cartesian space. Although the initial value of the error signal is a bit large due to the initial position, the error signal is totally kept to remain in the predefined constrained set.

Case 2 *(State Constraints in Free Space):* A new design is added to this case, which is involved to tackle with all state constraints. With all parameters being the same as those previously defined in Case 1, new parameters are added as: $K_2 = \text{diag}[10.0]$, the frontier of the constraint is $k_b = 1$.

The simulation results are given in Fig. 4 with the proposed controller (48). The tracking performance is plotted in Fig. 4(a), where the proposed control (48) has a capability to handle the effect of state constraints in free space. The error signals z_1 and z_2 are given in Fig. 4(b) and (c), which illustrates that error signals converge to predefined sets given that the initial conditions are bounded, and the tracking objective is achieved without the violation of constraint on z_1 and z_2. The control input is shown in Fig. 4(d). The motion of the end-effector in Cartesian space is given in Fig. 4(e), where the performance is smooth and the effect of state constraints are tackled by using the proposed control (48). Fig. 4(f) plots the error signal in Cartesian space. Although the initial value of error signal is a little big because of the initial position, the error signal has an ability to remain in the predefined constrained set.

Case 3 *(Output Constraint in Constrained Space):* In this case, the robot is required to move along the solid part of the circle in free space, and slide along the wall when in contact with a wall located at $x_{12} = 0.8$ m. The commanded trajectory is described as (59). Other parameters are the same as those previously defined in Case 1. In this case, we discuss first the effectiveness of the proposed control (23) with output constraint. It should be noted that the environment changes and external disturbances will degrade the system performance in the presence of constraints. To address such a problem, we consider the frontier of the output constraint is $k_a = [0.05, 0.3, 0.05]$, which does also make sense to verify the effectiveness of the proposed approach.

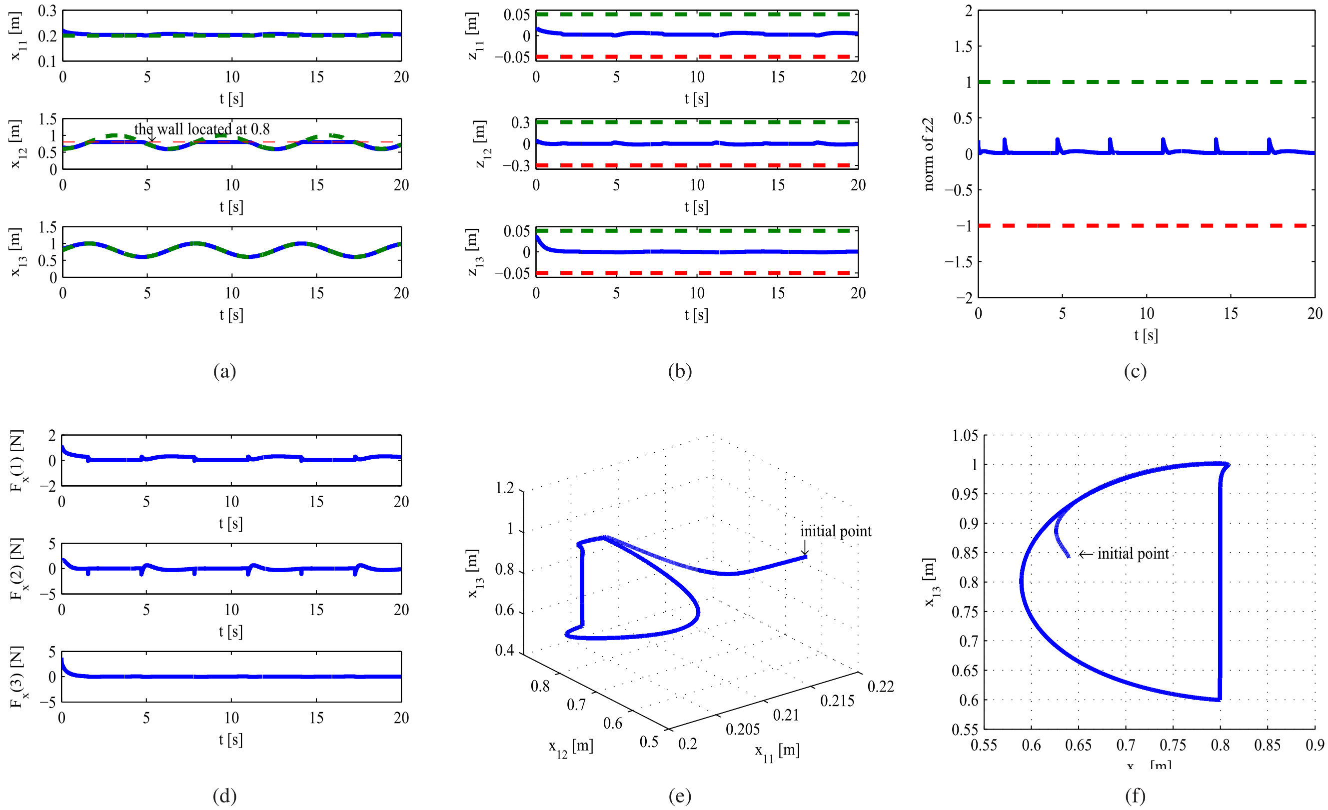

Fig. 6. Simulation results for Case 4: state constraints in constrained space. (a) x_1 (solid line: actual; dashed line: reference). (b) z_1 (solid line: actual; solid line: constraint of z_1). (c) z_2 (solid line: actual; solid line: constraint of z_2). (d) F_x. (e) Output signals in Cartesian space. (f) Output signals $x_{12} - x_{13}$ in Cartesian space.

Fig. 5(a) shows the position tracking when the end-effector is needed to tackle the change of the environment. The robot can track the commanded trajectory in free space, and slide along the wall when maintaining in contact with the wall. Fig. 5(b) plots the error signal z_1, which illustrates the error signals converge to a small value near zero and are always kept in the predefined constrained sets given that the initial condition is bounded. The control input is plotted in Fig. 5(c). It is noted that there are a few oscillations while the robot comes in contact with the wall. This is due to the change in the unknown environment, but the control force tends immediately to be smooth by using the proposed control (48). Fig. 5(d) shows that output signals are kept to remain in the constrained set. From Fig. 5(e), we can find that the value of x_{12} converges to $x_{12} = 0.8$ where the wall is located. Fig. 5(f) shows that the error signals are guaranteed to remain in constrained space.

Case 4 *(State Constraints in Constrained Space):* A new design to tackle all state constraints is added in this case with the proposed control (48). With all parameters being the same as those previously defined in Case 3. The new parameters are set as: $K_2 = \text{diag}[10.0]$, the frontier of the constraint is $k_b = 1$.

The tracking performance is shown in Fig. 6(a). It is clear that the robot tracks the commanded trajectory in free space, and slide along the wall when in contact phase. The error signal z_1 is plotted in Fig. 6(b). It is noticeable that the error signal z_1 converges to zero and is kept to the predefined constrained set. Another error signal z_2 is given in Fig. 6(c). Although there exist some much small oscillations due to the change of the environment, and z_2 can converge to zero immediately by using the impedance learning. The control input is given in Fig. 6(d). There are some oscillations of the control input when the robot contacts with the wall, because the designed control input requires large control effect while contacting with the wall to pull back the states to the constraint sets. Fig. 6(e) states that all state signals are kept to remain in constrained space. We can state that the value of x_{12} converges to $x_{12} = 0.8$ where the wall is located from Fig. 6(f).

V. CONCLUSION

Adaptive fuzzy NN control had been proposed for an uncertain constrained robot with unknown dynamics and constraints using impedance learning in this paper. Two control design approaches were investigated for constrained robots: 1) control design with output constraint and 2) control design with all state constraints. A *tan*-type BLF had been employed to handle the effect of constraint. The uncertain dynamics of robots had been approximated online by using the learning capability from the fuzzy NN structure. Four cases were investigated to verify the effectiveness of the proposed approach: 1) control with output constraint in free space; 2) control with all state constraints in free space; 3) control with output constraint in constrained space; and 4) control with all state constraints in constrained space. Simulation studies had been carried out to illustrate that the proposed control had a good performance to handle the robot–environment interaction.

APPENDIX A
PROOF FOR LEMMA 2

We first show that $\zeta(t)$ is bounded on $[0, t_f)$ by seeking a contradiction. To start, we rewrite (12) as

$$0 < V(t) \leq c_0 e^{-c_1 t} + e^{-c_1 t}\int_0^t g_1(t)N(\zeta)\dot{\zeta}e^{c_1\tau}d\tau$$
$$\leq c_0 e^{-c_1 t} + e^{-c_1 t}\int_0^t g_1(t)N(\zeta)\dot{\zeta}d\tau. \quad (60)$$

Depending on the sign of $\dot{\zeta}(t)$, two cases are considered.

Case 1: $\dot{\zeta}(t) \geq 0$. To deal with the term $\int_0^t g_1(t)N(\zeta)\dot{\zeta}d\tau$, we define the function $N_1(\zeta)$ by

$$N_1(\zeta) = \begin{cases} |l_1^+|N(\zeta), & \text{if } \zeta \in [4m-1, 4m+1) \\ |l_1^-|N(\zeta), & \text{if } \zeta \in [4m+1, 4m+3) \end{cases} \quad (61)$$

where l_1^- and l_1^+ are the lower and upper bounds of $g_1(t)$, and m is an integer. It is noted that the function $N(\zeta) = \zeta\cos((\pi\zeta/(2)))$ is positive on intervals $[4m-1, 4m+1)$ and negative on intervals $[4m+1, 4m+3)$ with m an integer, and $\dot{\zeta}(t) \geq 0$, (60) can be further written as

$$0 < V(t) \leq c_0 e^{-c_1 t} + e^{-c_1 t}\mathrm{sgn}(g_1)\int_0^{\zeta(t)} N_1(\zeta)d\zeta \quad \forall t \in [0, t_f). \quad (62)$$

In this case, suppose that $\zeta(t)$ has no upper bounds on $[0, t_f)$. The properties of Nussbaum-type function ensure the existence of two monotonely increasing sequences $\{\zeta_n^{(j)}\}$ with $\zeta_1^{(j)} > |\zeta(0)|$ and $\lim_{n\to\infty}\zeta_n^{(j)} = \infty$, $j = 1, 2$, such that

$$\lim_{n\to\infty}\frac{1}{\zeta_n^{(1)}}\int_0^{\zeta_n^{(1)}} N_1(s)ds = +\infty \quad (63)$$
$$\lim_{n\to\infty}\frac{1}{\zeta_n^{(2)}}\int_0^{\zeta_n^{(2)}} N_1(s)ds = -\infty. \quad (64)$$

Since $\zeta(t)$ has no upper bounds on $[0, t_f)$, therefore there are two monotonely increasing sequences $\{t_n^{(j)}\}$, $j = 1, 2$, such that $\zeta(t_n^{(j)}) = \zeta_n^{(j)}$, $j = 1, 2$. Clearly, $\lim_{n\to\infty} t_n^{(j)} = t_f$, $j = 1, 2$. Dividing (62) by $\zeta(t_n^{(j)}) = \zeta_n^{(j)} > 0$, there is

$$0 < \frac{V(t_n^{(1)})}{\zeta(t_n^{(1)})} \leq \frac{c_0 e^{-c_1 t_n^{(1)}}}{\zeta_n^{(1)}} + \frac{e^{-c_1 t}\mathrm{sgn}(g_1)}{\zeta_n^{(1)}}\int_0^{\zeta_n^{(1)}} N_1(\zeta)d\zeta \quad (65)$$

$$0 < \frac{V(t_n^{(2)})}{\zeta(t_n^{(2)})} \leq \frac{c_0 e^{-c_1 t_n^{(2)}}}{\zeta_n^{(2)}} + \frac{e^{-c_1 t}\mathrm{sgn}(g_1)}{\zeta_n^{(2)}}\int_0^{\zeta_n^{(2)}} N_1(\zeta)d\zeta \quad (66)$$

where $n = 1, 2, \ldots$. Please note that $t_n^{(j)} \to t_f$, $\zeta_n^{(j)} \to \infty$, $j = 1, 2$ when $n \to \infty$. Thus, if $\mathrm{sgn}(g_1) = 1$, (66) contradicts (64), and if $\mathrm{sgn}(g_1) = -1$, (65) contradicts (63). Therefore, $\zeta(t)$ is upper bounded on $[0, t_f)$.

Case 2: $\dot{\zeta}(t) \leq 0$. Similarly, there exists a contradiction no matter what the sign of $g_1(t)$ is. Therefore, $\zeta(t)$ is lower bounded on $[0, t_f)$.

Therefore, it can be concluded that the boundedness of $\zeta(t)$ on $[0, t_f)$. As an immediate result, $V(t)$ and $\int_0^t g_1(t)N(\zeta)\dot{\zeta}d\zeta$ are also bounded on $[0, t_f)$.

APPENDIX B
PROOF FOR THEOREM 1

Multiplying (37) by $e^{\rho_2 t}$ yields: $(d/(dt))(V_2(t)e^{\rho_2 t}) \leq c_2 e^{\rho_2 t}$. Integrating the above-mentioned inequality, we obtain $V_2 \leq e^{-\rho_2 t}V_2(0) + (c_2/(\rho_2)) - (c_2/(\rho_2))e^{-\rho_2 t} \leq V_2(0) + (c_2/(\rho_2))$. For z_1, we have $(1/(2))z_1^T z_1 \leq V_2(0) + (c_2/(\rho_2))$. Then, we can obtain $||z_1|| \leq \sqrt{B_1}$, where $B_1 = 2(V_2(0) + (c_2/(\rho_2)))$.

APPENDIX C
PROOF FOR THEOREM 2

We can rewritten (55) as $V_4(t) \leq e^{-\rho_4 t}\int_0^t g_1(\tau)N(z_2)\dot{z}_2 e^{\rho_4\tau}d\tau + e^{-\rho_4 t}(V_4(0) - (c_4/(\rho_4))) + (c_4/(\rho_4)) \leq V_4(0) + (c_4/(\rho_4)) + \int_0^t g_1(\tau)N(z_2)\dot{z}_2 d\tau$. According to Lemma 2, $\int_0^t g_1(\tau)N(z_2)\dot{z}_2 d\tau$ must be bounded on $[0, t_f)$, and $\| \int_0^t g_1(\tau)N(z_2)\dot{z}_2 d\tau\| \leq M$ with a constant $M > 0$. For z_1, we have $(1/(2))z_1^T z_1 \leq V_4(0) + (c_4/(\rho_4)) + M$. Then, we can obtain $||z_1|| \leq \sqrt{B_2}$, where $B_2 = 2(V_4(0) + (c_4/(\rho_4)) + M)$.

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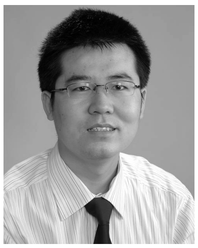

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